

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

# Two-variable data: Models and scatterplots

# 02

10 answers · Rationale-rich · Teach from doc

**Q1****B**

B. As the elevation increases, the temperature tends to decrease.

Choice B is correct. The association between the elevation and the temperature of locations in the Lake Tahoe Basin can be described by looking at the direction of the line of best fit. The line of best fit slopes downward, which corresponds to the temperature decreasing as the elevation increases.

Choices A and C are incorrect. Both of these choices would be represented by a line of best fit that slopes from the lower left to the upper right of the graph, which isn't what's shown on the graph. Choice D is incorrect. This choice would be represented by a line of best fit that is horizontal or has a slope very close to 0. This is not what's shown on the graph.

**Q2****A**

A. 2002

Choice A is correct. The year with the fewest graduates who enrolled in college within 24 months of graduation is the point with the lowest value on the vertical axis. This occurs at 2002.

Choice B, C, and D are incorrect. The years 2004, 2005, and 2007 each had a greater number of graduates who enrolled in college within 24 months of graduation than did the year 2002.

**Q3****480**

The correct answer is 480. An average temperature of  $32^{\circ}\text{C}$  corresponds to the value 32 on the x-axis. On the line of best fit, an x-value of 32 corresponds to a y-value of 480. The values on the y-axis correspond to the number of people predicted to visit this beach. Therefore, 480 people are predicted to visit this beach on a day with an average temperature of  $32^{\circ}\text{C}$ .

**Q4****B**

B.  $y = 2x^2 - 11x + 20$

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Choice B is correct. The graphical model that most closely fits the data in the scatterplot is a model in which the number of data points above and below the model are approximately balanced. Fitting a graphical model to the data shown results in an upward-facing parabola with a y-intercept near  $(0, 20)$  and a vertex with an approximate x-value of 2.5. Of the given choices, only choice B gives an equation of an upward-facing parabola with a y-intercept at  $(0, 20)$ . Furthermore, substituting 2.5 for x into the equation in choice B yields  $y = 5$ . This is approximately the y-value of the vertex of the model.

Choices A, C, and D are incorrect. These equations don't give a graphical model that best fits the data. At  $x = 0$ , they have y-values of  $-20$ ,  $-3$ , and 3, respectively. At  $x = 2.5$ , they have y-values of  $-35$ ,  $-3$ , and 3, respectively.

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**Q5****A**

A. Between 0 and 10 minutes

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Choice A is correct. The average rate of change in temperature of the coffee in degrees Fahrenheit per minute is calculated by dividing the difference between two recorded temperatures by the number of minutes in the corresponding interval of time. Since the time intervals given are all 10 minutes, the average rate of change is greatest for the points with the greatest difference in temperature. Of the choices, the greatest difference in temperature occurs between 0 and 10 minutes.

Choices B, C, and D are incorrect and may result from misinterpreting the average rate of change from the graph.

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**Q6****B**

B.  $y = 13.5 - 0.8x$

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Choice B is correct. The line of best fit shown intersects the  $y$ -axis at a positive  $y$ -value and has a negative slope. The graph of an equation of the form  $y = a + bx$ , where  $a$  and  $b$  are constants, intersects the  $y$ -axis at a  $y$ -value of  $a$  and has a slope of  $b$ . Of the given choices, only choice B represents a line that intersects the  $y$ -axis at a positive  $y$ -value, 13.5, and has a negative slope, -0.8.

Choice A is incorrect. This equation represents a line that has a positive slope, not a negative slope.

Choice C is incorrect. This equation represents a line that intersects the  $y$ -axis at a negative  $y$ -value, not a positive  $y$ -value, and has a positive slope, not a negative slope.

Choice D is incorrect. This equation represents a line that intersects the  $y$ -axis at a negative  $y$ -value, not a positive  $y$ -value.

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**Q7****C**

C. 6

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Choice C is correct. For each point on the scatterplot shown, the  $x$ -value represents the weight, in pounds, of a female gray wolf and the  $y$ -value represents the number of offspring that wolf produced. The point on the graph with an  $x$ -value of 50 has a  $y$ -value of 6. Therefore, the 50-pound gray wolf produced 6 offspring.

Choice A is incorrect. One of the wolves produced 8 offspring, but its weight was greater than 50 pounds.

Choice B is incorrect. Three of the wolves produced 7 offspring each, but their weights were each greater than 50 pounds.

Choice D is incorrect. Two of the wolves produced 5 offspring each, but their weights were each less than 50 pounds.

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**Q8****A**

A.  $y = -x + 6$

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Choice A is correct. A line of best fit for the data in a scatterplot is a line that follows the trend of the data with approximately half the data points above and half the data points below the line. Based on the given data, a line of best fit will have a positive y-intercept on or near the point  $(0, 6)$  and a negative slope. All of the choices are in slope-intercept form  $y = mx + b$ , where  $m$  is the slope and  $b$  is the y-coordinate of the y-intercept. Only choice A is an equation of a line with a positive y-intercept at  $(0, 6)$  and a negative slope,  $-1$ .

Choice B is incorrect. This equation is for a line that has a negative y-intercept, not a positive y-intercept. Choices C and D are incorrect and may result from one or more sign errors and from switching the values of the y-intercept and the slope in the equation.

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**Q9****B**

B. Theresa's speed was increasing for a longer period of time than it was decreasing.

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Choice B is correct. Theresa's speed was increasing from 0 to 5 minutes and from 20 to 25 minutes, which is a total of 10 minutes. Theresa's speed was decreasing from 10 minutes to 20 minutes and from 25 to 30 minutes, which is a total of 15 minutes. Therefore, Theresa's speed was NOT increasing for a longer period of time than it was decreasing.

Choice A is incorrect. Theresa ran at a constant speed for the 5-minute period from 5 to 10 minutes. Choice C is incorrect. Theresa's speed decreased at a constant rate during the last 5 minutes, which can be seen since the graph is linear during that time. Choice D is incorrect. Theresa's speed reached its maximum at 25 minutes, which is within the last 10 minutes.

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## Q10

B

### B. Increasing linear

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Choice B is correct. The given values show that as  $x$  increases,  $fx$  also increases, which means that  $f$  is an increasing function. Furthermore,  $fx$  increases at a constant rate of 1 for each increase of  $x$  by 1. A function with a constant rate of change is linear. Thus, the function  $f$  can be described as an increasing linear function.

Choice A is incorrect. For a decreasing linear function, as  $x$  increases,  $fx$  decreases rather than increases.

Choice C is incorrect. For a decreasing exponential function, for each increase of  $x$  by 1,  $fx$  decreases by a fixed percentage rather than increases at a constant rate.

Choice D is incorrect. For an increasing exponential function, for each increase of  $x$  by 1,  $fx$  increases by a fixed percentage rather than at a constant rate.

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### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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